

Unit 12, Lesson 9: Evaluating Real-World Problems Using the Order of Operations – Part 1

Benchmark

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

Learning Target

- ☐ I can solve real-world problems involving the order of operations with rational numbers, including exponents and radicals.

12.9.1 Warm-Up: FAILURE is an Option

Mark Cuban is an American billionaire entrepreneur. He is the owner of an NBA team, the Dallas Mavericks, and one of the investors on the reality show *Shark Tank*.

He has been quoted as saying, "I wouldn't be where I am now if I didn't fail ... a lot. The good, the bad, it's all part of the success equation."

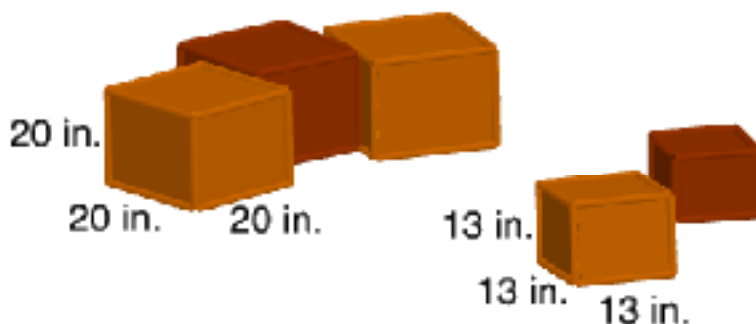
Failing can be when you try something that doesn't work out, but it can also be an opportunity.

1. Consider the following acronyms for F.A.I.L.
 - First Attempt In Learning
 - Forever Acquiring Important Lessons
 - First Action In Learning
- a. Star the acronym you connect with most.
- b. How can thinking about failure in this way be motivating?
- c. Create your own acronym to remind you that failure is just "part of the success equation."

12.9.2 Collaboration: Applying the Order of Operations for Expressions in Real-World Contexts

Work with your partner to complete the following.

1. Rodney needs to move some of his possessions into a storage unit. He has 3 cubic boxes that measure 20 inches (in.) on each side. He also has 2 cubic boxes that are 13 in. on each side.



To determine the amount of space all the boxes will take up, Rodney writes the following expression.

$$3 \cdot 20^3 + 2 \cdot 13^3$$

Rodney's calculations are shown.

$$3 \cdot 20^3 + 2 \cdot 13^3$$

$$3 \cdot 60 + 2 \cdot 39$$

$$180 + 78$$

$$258$$

- a. Draw an arrow to the step where Rodney first made an error.
- b. Correct Rodney's calculations in the space next to his work.

- c. How many cubic inches of storage does Rodney need?**
- d. Explain whether the smallest storage unit, which is 5 feet (ft) by 5 ft by 8 ft, will be large enough for his boxes.**

12.9.3 Guided Practice: Applying the Order of Operations Involving Exponents and Radicals for Real-World Contexts

- Doctors use a patient's Body Mass Index (BMI) as a diagnostic tool when analyzing their health. The following formula is used to calculate BMI, where w is a patient's weight in pounds (lb) and h is a patient's height in inches (in.).

$$\frac{w}{h^2} = 703$$

Determine the BMI for an adult who is 5 feet, 7 inches tall and weighs 159 pounds. Round to the nearest tenth.

- The surface area (SA) of a cone can be found using the formula $SA = \pi \cdot r \cdot \sqrt{r^2 + h^2}$, where r is the radius of the cone and h is the height of the cone. The expression shown can be used to find the surface area of a sugar cone with a radius of 2.25 centimeters (cm) and a height of 11 cm.

$$\pi \cdot 2.25 \cdot \sqrt{2.25^2 + 11^2}$$

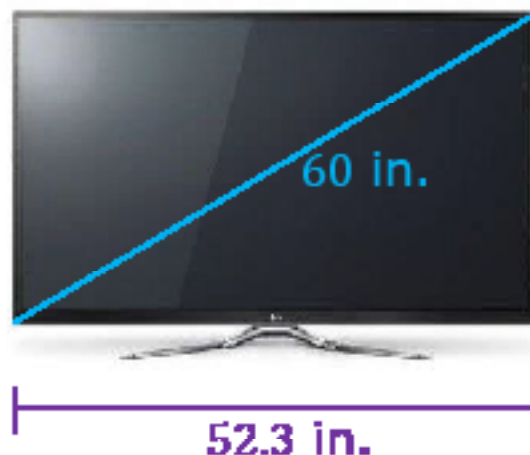
Find the surface area of the sugar cone in square cm. Use 3.14 to approximate π .



- Television screen dimensions are advertised using the measure of the diagonal distance across the screen. Kevin plans to buy a 60-inch television that measures 52.3 inches in width.

Kevin uses the expression shown to find the height of the space he needs to hang the television so there is at least 3 inches above and below the screen.

$$\sqrt{60^2 - 52.3^2} + 2(3)$$



Find the height of the space Kevin needs. Round to the nearest thousandth.

12.9.4 Your Turn!

Find the value of each expression. Show your work.

1. $\left(\frac{1}{5}\right)^2 \cdot (\sqrt{36} + 4)$

2. $(7^2 + \sqrt{25}) - 3.75$

3. Kinetic energy is the energy of an object in motion. It can be calculated using the expression $\frac{1}{2}mv^2$, where m is the mass of an object and v is the object's velocity.

If a cyclist has a mass of 71 kilograms (kg), his bicycle has a mass of 22 kg, and he is traveling at a speed of 8 meters (m) per second, the expression $\frac{1}{2}(71 + 22)(8)^2$ can be used to find the kinetic energy stored, measured in Joules.

Calculate the kinetic energy.

12.9.5 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on evaluating real-world problems using the order of operations?



